ATLAS-DEP-001

Deployment & Appliance Guide

Project: Project Atlas (Master Flightline Clock)
Version: 0.1 Draft
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1. Purpose

This guide defines how Atlas is deployed as a dedicated appliance. Atlas is designed to operate independently of the underlying operating system. Supported deployment targets include macOS and Microsoft Windows.

2. Design Philosophy

- Application logic remains platform independent.
- Operating system differences are isolated to deployment.
- The operator should never interact directly with the operating system during normal competition use.

3. Minimum Hardware

Component	Recommendation
Computer	Refurbished Mac Mini or Mini PC
Memory	8 GB minimum
Storage	128 GB SSD minimum
Display Outputs	Two displays (Operator + LED Output)
Operating System	macOS or Windows

4. Startup Sequence

1. System powers on.
2. Automatic user login.
3. Atlas Launcher starts automatically.
4. Launcher validates configuration, fonts and database.
5. Main Atlas application starts.
6. Operator Console and Public Display initialise.
7. System enters READY state.

5. Atlas Launcher Responsibilities

- Verify atlas.db exists.
- Create database if required.
- Validate application configuration.
- Record startup log.
- Launch Atlas.
- Optionally restart after abnormal termination.

6. Appliance Configuration

Requirement	Description
Automatic Login	Enabled
Sleep Mode	Disabled
Screen Saver	Disabled
Notifications	Disabled during operation
Automatic Updates	Deferred until maintenance periods

7. Administrative Functions

The Settings page shall provide an **Exit to Desktop** button. This function closes Atlas cleanly and returns the operator to the operating system desktop for maintenance tasks.

8. Hardware Profiles

Profile	Purpose
Development / LCD Monitor	Testing and development.
Single LED Panel (64×64)	Compact display layout.
Dual LED Panel (128×64)	Standard competition layout.

Hardware profile selection shall be available through the Settings page and stored within the Atlas database.

9. Future Enhancements

- Watchdog process.
- Automatic backup of database.
- Health monitoring.
- Remote diagnostics.

- Hardware communication abstraction layer.

10. Architectural Principle

Atlas shall be developed as a platform-independent Python appliance. Only deployment scripts differ between operating systems. All business logic, timing logic, database operations and rendering behaviour remain identical regardless of host platform.